

Practical No. 1 to 10 (Manual Testing Practicals)

Practical No.	Title	Tool Used
1	SDLC and STLC Phases	MS Word / Google Docs / Draw.io
2	Test Plan, RTM, Scope, Strategy	MS Word, MS Excel
3	Reviews, Walkthroughs, Inspections	MS Word, Sample SRS Document
4	Functional & Non-Functional Test Cases	MS Excel / Google Sheets
5	Defect Identification & Bug Reports	MS Excel / Bugzilla (Optional)
6	Manual Test Execution	MS Excel
7	BVA & Equivalence Partitioning	MS Excel / MS Word
8	Defect Management & Test Management	Bugzilla, TestLink
9	Black Box, White Box, Integration & System Testing	MS Word
10	Automation & Regression Testing Analysis	MS Word / Excel

Practical No. 11 to 15 (Python + Selenium Automation Testing)

Required Software

1. Python

Install Python 3.10 or later

Official Website:

[Python Official Website](https://www.python.org/)

Verify Installation:

```
python --version
```

2. Visual Studio Code (VS Code)

IDE for writing Selenium programs.

[Visual Studio Code](https://code.visualstudio.com/)

3. Selenium Library

Install using command:

```
pip install selenium
```

Verify:

```
pip show selenium
```

4. Google Chrome Browser

Required for browser automation.

[Google Chrome](https://www.google.com/chrome/)

5. ChromeDriver

(For older Selenium versions)

Download version matching Chrome browser.

[Chrome for Testing / ChromeDriver Downloads](https://chromedriver.chromium.org/downloads)

Note: Selenium 4.6+ automatically manages drivers using Selenium Manager, so manual ChromeDriver installation is often not required.

Tool Mapping for Practical No. 11–15

Practical	Tool Used
11	Python + Selenium + Chrome Browser + VS Code
12	ChatGPT, Gemini, Claude, Copilot (AI Tools)

13	Python + Selenium + VS Code
14	Python + Selenium WebDriver
15	Python + Selenium WebDriver + Assertions

Recommended Lab Setup (Per Student PC)

Software List

- ✓ Windows 10/11
- ✓ Python 3.10+
- ✓ VS Code
- ✓ Selenium Package
- ✓ Google Chrome
- ✓ Internet Connection
- ✓ Sample Testing Website

Sample Websites for Selenium Practicals

Form Handling

[DemoQA Practice Forms](#)

Login Automation

[SauceDemo Test Website](#)

Navigation Testing

[Wikipedia](#)

Web Elements Testing

[The Internet Test Site](#)

Selenium Setup Using VS Code

Step 1: Install Python

1. Download Python (3.10 or later) from <https://www.python.org/downloads/>
2. During installation, select **Add Python to PATH**.
3. Click **Install Now**.

Verify Installation

Open Command Prompt:

```
python --version
```

Step 2: Install Visual Studio Code (VS Code)

1. Download VS Code from <https://code.visualstudio.com/>

2. Install with default settings.
3. Open VS Code.

Step 3: Install Python Extension in VS Code

1. Open VS Code.
2. Go to **Extensions** (Ctrl+Shift+X).
3. Search for:
 - **Python** (Microsoft)
 - **Pylance**
4. Click **Install**.

Step 4: Create Project Folder

Create a folder:

Selenium_Project

Open this folder in VS Code.

Step 5: Open Terminal in VS Code

Select:

Terminal → New Terminal

Step 6: Install Selenium

Execute:

```
pip install selenium
```

Step 7: Verify Selenium Installation

```
pip show selenium
```

Expected output:

Name: selenium

Version: 4.x.x

Step 8: Create Python File

Create:

test.py

Step 9: Write Test Program

```
from selenium import webdriver  
driver = webdriver.Chrome()  
driver.get("https://www.google.com")  
print(driver.title)  
driver.quit()
```

Step 10: Execute Program

In VS Code Terminal:

```
python test.py
```

Expected Result

- Chrome browser opens automatically.
 - Google homepage loads.
 - Title "Google" is printed.
 - Browser closes automatically.
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Selenium Setup Using PyCharm

Step 1: Install Python

Download Python (3.10+) from:

<https://www.python.org/downloads/>

Verify:

```
python --version
```

Step 2: Install PyCharm Community Edition

Download from:

<https://www.jetbrains.com/pycharm/download/>

Install with default settings.

Step 3: Create New Project

1. Open PyCharm.
2. Click **New Project**.
3. Choose project location.
4. Select the Python interpreter.
5. Click **Create**.

Step 4: Install Selenium Package

Method 1: Using Terminal

Open Terminal in PyCharm and execute:

```
pip install selenium
```

Method 2: Using Package Manager

1. Go to:

File → Settings → Project → Python Interpreter

2. Click + (**Install Package**).
3. Search **selenium**.
4. Click **Install Package**.

Step 5: Verify Installation

Open Terminal:

```
pip show selenium
```

Step 6: Create Python File

Create:

```
test.py
```

Step 7: Write Program

```
from selenium import webdriver
```

```
driver = webdriver.Chrome()
```

```
driver.get("https://www.google.com")
```

```
print(driver.title)
```

```
driver.quit()
```

Step 8: Run Program

Right-click **test.py** and select:

Run 'test'

or execute:

```
python test.py
```

Testing and Verification Steps

Test Step	Expected Result
Python installed	Version displayed
VS Code/PyCharm installed	IDE opens successfully
Selenium installed	Package installed without error
pip show selenium executed	Selenium details displayed
test.py executed	Chrome browser opens

Google website loads	Successful navigation
Page title displayed	"Google" printed
Browser closes	Program ends successfully

Note

For **Practical 1–10**, MS Word and MS Excel are sufficient because they are documentation and analysis-based practicals.

For **Practical 11–15**, students must use **Python + Selenium WebDriver + VS Code + Chrome Browser** to execute and capture screenshots of the automation scripts.